

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

14

Q1**C**

C. $16512.7 < x < 16515.7$

Choice C is correct. It's given that the estimated volume of the slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet. It's also given that the estimated weight per cubic foot of volume of that slab is 165 pounds. The estimated total weight of the slab of granite, in pounds, can be calculated by multiplying the estimated volume by the estimated weight per cubic foot. Therefore, if the estimated volume of the slab of granite is 12.7 cubic feet, its estimated total weight is $165(12.7)$ pounds, and if the estimated volume of the slab of granite is 15.7 cubic feet, its estimated total weight is $165(15.7)$ pounds. Since the estimated volume of the slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet, the estimated total weight x , in pounds, must be greater than $165(12.7)$ pounds and less than $165(15.7)$ pounds. This situation can be represented by the inequality $165(12.7) < x < 165(15.7)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**A**

A. $y = 6x + 8$

Choice A is correct. The slope-intercept form of an equation for a line is $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept of the line. It's given that the slope is 6, so $m = 6$. It's also given that the line passes through the point $(0, 8)$ on the y-axis, so $b = 8$. Substituting $m = 6$ and $b = 8$ into the equation $y = mx + b$ gives $y = 6x + 8$.

Choices B, C, and D are incorrect and may result from misinterpreting the slope-intercept form of an equation of a line.

Q3**54**

The correct answer is 54. Dividing both sides of the given equation by 2 yields $x = 6$. Multiplying both sides of this equation by 9 yields $9x = 54$. Thus, the value of $9x$ is 54.

Q4**B**

B. $f(x) = \frac{3}{4}x + 3$

Choice B is correct. The equation for the function f in the xy -plane can be represented by $f(x) = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept. Since it's given that the line has a slope of $\frac{3}{4}$, it follows that $m = \frac{3}{4}$ in $f(x) = mx + b$, which yields $y = \frac{3}{4}x + b$. It's given that $f(0) = 3$. This implies that the graph of the function f in the xy -plane passes through the point $(0, 3)$. Thus, the y -coordinate of the y -intercept of the graph is 3, so $b = 3$ in $f(x) = \frac{3}{4}x + b$, which yields $f(x) = \frac{3}{4}x + 3$. Therefore, the equation $f(x) = \frac{3}{4}x + 3$ defines the function f .

Choice A is incorrect and may result from a sign error for the y -intercept. Choice C is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$, in addition to a sign error for the y -intercept. Choice D is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$.

Q5**B****B. 3**

Choice B is correct. Let n be the number of novels and m be the number of magazines that Sadie purchased. If Sadie purchased a total of 11 novels and magazines, then $n + m = 11$. It is given that the combined price of 11 novels and magazines is \$20. Since each novel sells for \$4 and each magazine sells for \$1, it follows that $4n + m = 20$. So the system of equations below must hold.

$$4n + m = 20$$

$$n + m = 11$$

Subtracting corresponding sides of the second equation from the first equation yields $3n = 9$, so $n = 3$. Therefore, Sadie purchased 3 novels.

Choice A is incorrect. If 2 novels were purchased, then a total of \$8 was spent on novels. That leaves \$12 to be spent on magazines, which means that 12 magazines would have been purchased. However, Sadie purchased a total of 11 novels and magazines. Choices C and D are incorrect. If 4 novels were purchased, then a total of \$16 was spent on novels. That leaves \$4 to be spent on magazines, which means that 4 magazines would have been purchased. By the same logic, if Sadie purchased 5 novels, she would have no money at all (\$0) to buy magazines. However, Sadie purchased a total of 11 novels and magazines.

Q6**B****B. 3**

Choice B is correct. It's given that the equation $y = 0.1x$ models the relationship between the number of different pieces of music a certain pianist practices, y , and the number of minutes in a practice session, x . Since it's given that the session lasted 30 minutes, the number of pieces the pianist practiced can be found by substituting 30 for x in the given equation, which yields $y = 0.1(30)$, or $y = 3$.

Choices A and C are incorrect and may result from misinterpreting the values in the equation. Choice D is incorrect. This is the given value of x , not the value of y .

Q7**D**

D. $2x + 3 = 6$

Choice D is correct. Dividing each side of the original equation by 2 yields $\frac{4x+6}{2} = \frac{12}{2}$, which simplifies to $2x + 3 = 6$.

Choice A is incorrect. Dividing each side of the original equation by 2 gives $2x + 3 = 6$, which is not equivalent to $2x + 4 = 6$. Choice B is incorrect. Dividing each side of the original equation by 4 gives $x + \frac{3}{2} = 3$, which is not equivalent to $x + 3 = 3$. Choice C is incorrect. Dividing each side of the original equation by 3 gives $\frac{4}{3}x + 2 = 4$, which is not equivalent to $3x + 2 = 4$.

Q8**A**

A. 2

Choice A is correct. Substituting 63 for $f(x)$ in the given function yields $63 = 9(2x + 3)$. Dividing both sides of this equation by 9 yields $7 = 2x + 3$. Subtracting 3 from both sides of this equation yields $4 = 2x$. Dividing both sides of this equation by 2 yields $2 = x$. Therefore, $f(x) = 63$ when the value of x is 2.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**A**

A. $\begin{cases} x + y = 120 \\ 90x + 250y = 27,600 \end{cases}$

Choice A is correct. The airline sold two types of tickets for this flight: x tickets at \$90 each and the remaining tickets, y , at \$250 each. Because the airline sold a total of 120 tickets for this flight, it must be true that $x + y = 120$. The amount, in dollars, collected from the sale of x tickets at \$90 each is represented by $90x$. The amount, in dollars, collected from the sale of the remaining y tickets at \$250 each is represented by $250y$. It is given that a total of \$27,600 was collected from the sale of all tickets. Therefore, it must also be true that $90x + 250y = 27,600$.

Choice B is incorrect. The total number of tickets sold is represented correctly as $x + y = 120$. The total amount, in dollars, collected from the sale of the x tickets at \$90 each and the remaining tickets, y , at \$250 has been correctly represented as $90x + 250y$. However, according to the information given, this total should be equal to 27,600, not $120(27,600)$ dollars. Choice C is incorrect. The total number of tickets sold has been correctly represented as $x + y$. However, according to the information given, this total should be equal to 120, not 27,600, as shown in choice C. The total amount, in dollars, collected from the sale of the x tickets at \$90 each and the remaining tickets, y , at \$250 has been correctly represented as $90x + 250y$. However, according to the information given, this total should be equal to 27,600, not $120(27,600)$ dollars. Choice D is incorrect. The two equations given in choice D have no meaning in this context.

Q10

C

C. 100

Choice C is correct. Let d represent the mass, in grams, of vitamin D in the mixture, and let c represent the mass, in grams, of calcium in the mixture. It's given that the mixture consists of only vitamin D and calcium and that the total mass of the mixture is 150 grams. Therefore, the equation $d + c = 150$ represents this situation. It's also given that the mass of vitamin D in the mixture is 50 grams. Substituting 50 for d in the equation $d + c = 150$ yields $50 + c = 150$. Subtracting 50 from both sides of this equation yields $c = 100$. Therefore, the mass of calcium in the mixture is 100 grams.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the total mass, in grams, of the mixture, not the mass, in grams, of calcium in the mixture.

Choice D is incorrect. This is the mass, in grams, of vitamin D in the mixture, not the mass, in grams, of calcium in the mixture.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank